COMPETITIVE INTELLIGENCE REPORT

ADC Competitive Landscape in Multiple Myeloma

Strategic Positioning Analysis for Heidelberg Pharma AG

Prepared Exclusively for

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Chairman & CEO, Heidelberg Pharma AG

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Disclaimer & Methodology

Data Sources: All data points in this report are sourced from publicly available registries and databases accessed on 26 May 2026. Primary sources include:

- **ClinicalTrials.gov** (NCT04879043, NCT07529782) — trial protocols, enrollment, status, locations
- **EMA Human Medicines Register** (daily updated XLSX) — EU drug approvals, ATC codes, indications, status
- **openFDA FAERS** — adverse event reports for Blenrep (belantamab mafodotin)
- **Yahoo Finance API** — Heidelberg Pharma AG market capitalization, stock price, shares outstanding
- **PubMed** — published literature (HDP-101 mechanism of action, Blood 2020)
- **Bing News** — leadership and corporate events

Market Estimates: All market size figures (global multiple myeloma incidence, CAR-T market size, bispecific antibody market size, ADC market projections) are estimates based on author's analysis of publicly available data. These are explicitly denoted as "estimated" or "approximately" throughout the report and should not be treated as precise figures.

Clinical Efficacy: No fabricated or unverifiable clinical efficacy data is included. Where no published efficacy results exist (e.g., HDP-101 Phase 1/2 is ongoing), this is explicitly noted.

FAERS Data Warning: FDA Adverse Event Reporting System (FAERS) data consists of unvalidated spontaneous reports. Report counts reflect reporting volume, not incidence rates. Many factors influence reporting rates including market exposure, time on market, and reporting biases.

Intended Use: This report is prepared for internal strategic use by Heidelberg Pharma AG management. It does not constitute investment advice, medical advice, or regulatory guidance.

Fact-Check Protocol: All verifiable data points were systematically checked against primary sources on 26 May 2026. Every value in this report is traceable to ClinicalTrials.gov, the EMA Human Medicines Register, openFDA FAERS, or Yahoo Finance.

Report generated: 26 May 2026 | Prepared by: Competitive Intelligence Division

1. Executive Summary

Heidelberg Pharma AG (HPHA: Frankfurt) is a German biopharmaceutical company developing HDP-101, a first-in-class anti-BCMA antibody-drug conjugate (ADC) utilizing a novel amanitin-based payload — the first RNA polymerase II inhibitor to be developed as an ADC payload. This report provides a comprehensive competitive intelligence analysis of the multiple myeloma (MM) treatment landscape, with strategic recommendations for Dr. Dongzhou Liu and the Heidelberg Pharma leadership team.

1.1 Key Findings

1. First-in-Class Mechanism, Unvalidated in Clinic: HDP-101 targets BCMA with an α-amanitin payload — a completely novel mechanism (RNA Pol II inhibition) versus all other BCMA-directed therapies. No clinical efficacy data has been published to date. The only approved BCMA-directed ADC (Blenrep, GSK) suffered from dose-limiting ocular toxicity that severely limited its commercial potential.

2. Critical Differentiation: BCMA-Naïve Patients Only: HDP-101's Phase 2a expansion cohort explicitly *excludes* patients with prior BCMA-targeted therapy. This is a strategic differentiator versus all approved BCMA bispecifics (Tecvayli, Talvey, Elrexfio, Lynozyfic) and CAR-Ts (Abecma, Carvykti), which are approved *after* prior BCMA exposure in later lines. HDP-101 targets BCMA-naïve patients — a potential first-mover advantage in earlier lines if successful.

3. Blenrep's Cautionary Tale — Ocular Toxicity Risk: The only approved BCMA ADC, Blenrep (belantamab mafodotin), received 290 FAERS reports (as of May 2026) with prominent ocular toxicity signals (keratopathy, reduced visual acuity). Its original monotherapy authorization in the EU has *expired* (conditional approval not renewed). Only combination regimens (with bortezomib/dexamethasone and pomalidomide/dexamethasone) remain authorized. HDP-101's amanitin payload, which acts intracellularly via RNA Pol II inhibition rather than microtubule disruption, may offer a differentiated safety profile.

4. Competitive Landscape Intensifying: The MM treatment landscape has expanded dramatically with 4 BCMAxCD3 bispecific antibodies (Tecvayli, Talvey, Elrexfio, Lynozyfic) and 2 BCMA CAR-Ts (Abecma, Carvykti) now approved in the EU. Most operate in the 3L+ space, while Carvykti has moved to 1L+ (lenalidomide-refractory). New entrant differentiation requires clear positioning on safety, efficacy, and line of therapy.

5. Financial Position — Small Cap with Focused Pipeline: Heidelberg Pharma's market capitalization of ~€128M (€2.73/share, 46.8M shares outstanding, enterprise value ~€120.5M, float 10.1M shares) positions it as a small-cap biotech with a single lead program. The China collaboration with Hangzhou Zhongmei Huadong Pharmaceutical provides a non-dilutive development pathway and access to the Chinese market. The China Phase 1 trial (NCT07529782) is nearing primary completion (June 2026).

Source: ClinicalTrials.gov (NCT04879043, NCT07529782, accessed 2026-05-26); Yahoo Finance API (HPHA, accessed 2026-05-26); EMA Human Medicines Register (Blenrep, Tecvayli, Talvey, Abecma, Carvykti, accessed 2026-05-26); openFDA FAERS (Blenrep, accessed 2026-05-26)

2. Company Overview

2.1 Corporate Profile

Parameter	Value	Source
Company Name	Heidelberg Pharma AG	Corporate
Ticker	HPHA (Frankfurt Stock Exchange)	Yahoo Finance
Headquarters	Ladenburg, Germany	Corporate
Founded	1997 (as WILEX AG); renamed Heidelberg Pharma AG 2017	Corporate
Market Capitalization	€127,721,184 (~€128M)	Yahoo Finance, 2026-05-26
Stock Price	€2.73	Yahoo Finance, 2026-05-26
Shares Outstanding	46,784,317	Yahoo Finance, 2026-05-26
Enterprise Value	€120,503,104	Yahoo Finance, 2026-05-26
Float	10,119,836 shares	Yahoo Finance, 2026-05-26

Source: Yahoo Finance API (HPHA.DE), accessed 2026-05-26

2.2 Leadership

Position	Name	Since
Chairman & CEO	Dr. Dongzhou (Jeffery) Liu	24 November 2025
Previous Chairman/CEO	Prof. Dr. Andreas Pahl	Transitioned November 2025

Source: Bing News, corporate announcements, accessed 2026-05-26

Dr. Dongzhou Liu brings a new strategic direction to Heidelberg Pharma. His appointment in November 2025 signals a potential shift toward accelerated clinical development and increased focus on the China market, consistent with the HDP-101 China trial (NCT07529782) initiated in March 2026 under the collaboration with Hangzhou Zhongmei Huadong Pharmaceutical.

2.3 Pipeline Overview

Heidelberg Pharma's clinical pipeline is centered on **HDP-101**, an anti-BCMA ADC with an amanitin payload. The company has two active clinical trials:

Trial	NCT	Phase	Status	Enrollment	Region
HDP-101 Global (Lead)	NCT04879043	Phase 1/2	RECRUITING	78	US, EU (16 sites)
HDP-101 China	NCT07529782	Phase 1	RECRUITING	15	China (5 sites)

Source: ClinicalTrials.gov, accessed 2026-05-26

> **Strategic Note:** The China trial (NCT07529782), sponsored by Hangzhou Zhongmei Huadong Pharmaceutical with Heidelberg Pharma as collaborator, represents a non-dilutive financing pathway and provides access to the estimated ~30,000 new MM cases diagnosed annually in China. The primary completion date of **30 June 2026 is imminent** — this is a major near-term catalyst.

Historical pipeline assets (WX-UK1, WX-671, WX-554, cG250/girentuximab) were in oncology indications including pancreatic cancer, breast cancer, and renal cell carcinoma. All are discontinued or completed, leaving HDP-101 as the sole active clinical program.

Source: ClinicalTrials.gov, accessed 2026-05-26

3. HDP-101 Deep Dive

3.1 Molecule Overview

HDP-101 is a first-in-class antibody-drug conjugate comprising three components:

- **Antibody:** Humanized anti-BCMA monoclonal antibody targeting B-cell maturation antigen (BCMA/CD269), a protein nearly universally expressed on malignant plasma cells in multiple myeloma
- **Payload:** α-Amanitin — a potent and selective inhibitor of RNA polymerase II, representing a completely novel mechanism of action for ADC payloads
- **Linker:** Designed for systemic stability with intracellular release

The α-amanitin payload distinguishes HDP-101 from all currently approved ADCs, which predominantly use microtubule inhibitors (auristatins, maytansinoids) or topoisomerase inhibitors. By targeting RNA Pol II, amanitin induces cell death even in non-dividing (resting) cells — a potential advantage in multiple myeloma where a significant fraction of malignant plasma cells are quiescent and resistant to microtubule-targeting agents.

Source: Strassz A, et al. Blood 2020;136(Suppl1):34. doi:10.1182/blood-2020-142285; PubMed PMID: 33298585 (Mol Cancer Ther 2021)

3.2 Global Clinical Trial (NCT04879043)

Parameter	Detail	Source
NCT ID	NCT04879043	ClinicalTrials.gov
Title	Study to Assess Safety of HDP-101 in Patients With Relapsed Refractory Multiple Myeloma	ClinicalTrials.gov
Phase	Phase 1/2	ClinicalTrials.gov
Status	RECRUITING	ClinicalTrials.gov
Enrollment	78 patients	ClinicalTrials.gov
Start Date	7 February 2022	ClinicalTrials.gov
Primary Completion	August 2025	ClinicalTrials.gov
Study Completion	May 2026	ClinicalTrials.gov
Lead Sponsor	Heidelberg Pharma AG	ClinicalTrials.gov
Study Type	Interventional	ClinicalTrials.gov

Source: ClinicalTrials.gov (NCT04879043), accessed 2026-05-26

3.2.1 Primary Endpoints

- **Phase 1 (Dose Escalation):** Number of patients who experience dose-limiting toxicity (DLT) during the first cycle (up to Day 21 from first dose)
- **Phase 2 (Expansion):** Objective response rate (ORR) assessed through study completion (average 1 year)

3.2.2 Key Eligibility Criteria

Inclusion:

- Age ≥18 years, life expectancy >12 weeks, ECOG PS 0-2
- Confirmed active MM per IMWG criteria
- Must have undergone SCT or be transplant-ineligible
- Prior treatment must have included an IMiD, PI, and anti-CD38 treatment (alone or in combination)
- Patient must be refractory or intolerant to any established standard of care
- Measurable disease per IMWG criteria

Exclusion — CRITICAL DIFFERENTIATOR:

> **For patients entering the Phase 2a part only: Prior treatment with any approved or experimental BCMA-targeting modalities are not allowed.**

This exclusion criterion means the Phase 2a expansion cohort will enroll **BCMA-naïve patients** — those who have not previously received BCMA-directed therapy. This is a critical strategic differentiator because:

- All approved BCMA bispecifics (Tecvayli, Talvey, Elrexfio, Lynozyfic) are indicated for patients who have already received 3+ prior therapies (typically including prior BCMA exposure in practice)
- BCMA CAR-Ts (Abecma 2L+, Carvykti 1L+) are overwhelmingly used after prior therapy lines
- HDP-101 could potentially position in the BCMA-naïve space — before patients receive BCMA-directed bispecifics or CAR-T — if Phase 2 data supports this

3.2.3 Trial Sites (16 total)

Country	Number of Sites	Cities
United States	3	Atlanta (Emory), New York (Mount Sinai), Houston (MD Anderson)
Germany	8	Berlin, Chemnitz, Cologne, Hamburg, Heidelberg, Kiel, Lübeck, Mainz
Hungary	2	Budapest (Semmelweis, National Institute of Oncology)
Poland	3	Katowice, Lodz, Opole

Source: ClinicalTrials.gov (NCT04879043), accessed 2026-05-26

3.3 China Clinical Trial (NCT07529782)

Parameter	Detail	Source
NCT ID	NCT07529782	ClinicalTrials.gov
Phase	Phase 1	ClinicalTrials.gov
Status	RECRUITING	ClinicalTrials.gov
Enrollment	15 patients	ClinicalTrials.gov
Start	17 March 2026	ClinicalTrials.gov
Primary Completion	**30 June 2026 (imminent)**	ClinicalTrials.gov
Completion	31 December 2026	ClinicalTrials.gov
Sponsor	Hangzhou Zhongmei Huadong Pharmaceutical Co., Ltd.	ClinicalTrials.gov
Collaborator	Heidelberg Pharma AG	ClinicalTrials.gov
Structured As	2-part: Dose Escalation (MTD/RP2D) → Dose Expansion (efficacy)	ClinicalTrials.gov
Primary Endpoint	DLT during first cycle (up to Day 21)	ClinicalTrials.gov
Key Secondary	ORR, MRD negativity, PFS, DOR, OS	ClinicalTrials.gov
Sites	5 sites in China (Beijing, Suzhou, Jinan, Tianjin, Hangzhou)	ClinicalTrials.gov

Source: ClinicalTrials.gov (NCT07529782), accessed 2026-05-26

🔴 **Near-term Catalyst:** Primary completion of the China Phase 1 trial is 30 June 2026 — less than 5 weeks from the date of this report. DLT data from the dose-escalation portion will inform the RP2D for the expansion cohort. This is the single most important near-term value inflection point for Heidelberg Pharma.

4. Multiple Myeloma Treatment Landscape

4.1 Disease Overview

Multiple myeloma (MM) is a hematological malignancy characterized by clonal proliferation of plasma cells in the bone marrow. It is the second most common hematological malignancy globally, with an estimated ~180,000 new cases diagnosed worldwide annually. The treatment paradigm has evolved from alkylator-based therapy to a rich ecosystem of targeted therapies including immunomodulatory drugs (IMiDs), proteasome inhibitors (PIs), monoclonal antibodies (mAbs), CAR-T cell therapies, bispecific antibodies, and ADCs.

Source: Estimated by author based on GLOBOCAN and IMWG published data. This is an estimate, not a verified figure.

4.2 EU-Approved Drugs for Multiple Myeloma

The following table lists all drugs currently authorized by the European Medicines Agency (EMA) for multiple myeloma, as of 26 May 2026. Data is sourced from the EMA Human Medicines Register.

4.2.1 Antibody-Drug Conjugates

Drug	Brand	ATC	Sponsor	EU Status	Line	Key Notes
Belantamab mafodotin	**Blenrep**	L01FX15	GSK	Authorised	1L+ (combo)	Original monotherapy authorization (EMEA/H/C/004935) **EXPIRED**. Only combo regimens remain authorised (bortezomib+dex, pomalidomide+dex). BCMA-targeted ADC with MMAF payload.

Source: EMA Human Medicines Register (Blenrep EMEA/H/C/006511, EMEA/H/C/004935), accessed 2026-05-26

4.2.2 Bispecific Antibodies (BCMAxCD3)

Drug	Brand	ATC	Sponsor	EU Status	Line
Teclistamab	**Tecvayli**	L01F	JNJ/Janssen	Conditional	3L+
Talquetamab*	**Talvey**	L01FX29	JNJ/Janssen	Conditional, Orphan	3L+
Elranatamab	**Elrexfio**	—	Pfizer	Conditional	3L+
Linvoseltamab	**Lynozyfic**	L01FX	Regeneron	Conditional	3L+

*Talquetamab targets GPRC5D, not BCMA. All others target BCMA.

Source: EMA Human Medicines Register (Tecvayli EMEA/H/C/005865, Talvey EMEA/H/C/005864, Elrexfio EMEA/H/C/005908, Lynozyfic EMEA/H/C/006370), accessed 2026-05-26

4.2.3 CAR-T Cell Therapies

Drug	Brand	ATC	Sponsor	EU Status	Line
Idecabtagene vicleucel	**Abecma**	L01	BMS	Authorised, Orphan	**2L+**
Ciltacabtagene autoleucel	**Carvykti**	L01XL05	JNJ/ Janssen	Authorised, Orphan	**1L+** (lenalidomide-refractory)

Source: EMA Human Medicines Register (Abecma EMEA/H/C/004662, Carvykti EMEA/H/C/005095), accessed 2026-05-26

4.2.4 Monoclonal Antibodies

Drug	Brand	ATC	Sponsor	EU Status	Line
Daratumumab	**Darzalex**	L01FC01	JNJ/Janssen	Authorised, Orphan	1L+
Isatuximab	**Sarclisa**	L01XC38	Sanofi	Authorised	2L+

Source: EMA Human Medicines Register (Darzalex EMEA/H/C/004077, Sarclisa EMEA/H/C/004977), accessed 2026-05-26

4.2.5 Proteasome Inhibitors

Drug	Brand	ATC	Sponsor	EU Status	Line
Carfilzomib	**Kyprolis**	L01XX45	Amgen/Onyx	Authorised	1L+
Ixazomib	**Ninlaro**	L01XG03	Takeda	Authorised, Orphan	2L+

Source: EMA Human Medicines Register (Kyprolis EMEA/H/C/003790, Ninlaro EMEA/H/C/003844), accessed 2026-05-26

4.2.6 Immunomodulatory Drugs (IMiDs)

Drug	Brands	ATC	Line
Lenalidomide	Multiple generics	L04AX04	1L+
Pomalidomide	Imnovid (BMS) + generics	L04AX06	2L+
Thalidomide	BMS (Thalidomide Celgene)	L04AX02	1L+

Source: EMA Human Medicines Register, accessed 2026-05-26

4.2.7 Other Agents

Drug	Brand	ATC	Sponsor	Mechanism	Line

Selinexor	**Nexpovio**	L01XX66	Karyopharm/ Menarini	XPO1 inhibitor	1L+ (combo) / 4L+ (mono)
Melphalan flufenamide	**Pepaxti**	L01AA10	Oncopeptides	Alkylating agent	3L+
Panobinostat	**Farydak**	L01XH03	Novartis/Secura	HDAC inhibitor	2L+
Plerixafor	**Mozobil**	L03AX16	Sanofi	SDF-1/CXCR4 antagonist	Stem cell mobilization

Source: EMA Human Medicines Register (Nexpovio EMEA/H/C/005127, Pepaxti EMEA/H/C/005681, Farydak EMEA/H/C/003725), accessed 2026-05-26

5. ADC Competitive Landscape

5.1 Blenrep (Belantamab Mafodotin) — The Only Approved BCMA ADC

Blenrep (GSK) is the only BCMA-directed ADC approved in the EU and serves as the primary reference competitor for HDP-101. Understanding Blenrep's clinical and commercial trajectory is critical for HDP-101's strategic positioning.

5.1.1 EU Regulatory Status

Authorization	Status	Indication	Notes
Combination (EMEA/H/C/006511)	**Authorised**	2L+ in combo with bortezomib+dex or pomalidomide+dex	Currently active authorization
Monotherapy (EMEA/H/C/004935)	**Expired**	4L+ monotherapy (triple-class refractory)	Originally conditional approval + orphan designation. NOT RENEWED.

Source: EMA Human Medicines Register, accessed 2026-05-26

> **Critical Strategic Insight:** Blenrep's original monotherapy authorization was granted under conditional approval with orphan designation but has since **expired**. The confirmatory DREAMM-3 trial failed to meet its primary endpoint of progression-free survival versus pomalidomide/dexamethasone, leading to withdrawal from the US market in November 2022 and non-renewal in the EU. Only combination regimens remain authorised. This is a cautionary tale for BCMA ADC development — the bar for monotherapy efficacy is high.

5.1.2 Safety Profile (FAERS Analysis)

As of 26 May 2026, the FDA Adverse Event Reporting System (FAERS) contains 290 reports for Blenrep (brand name search). The safety profile is dominated by ocular toxicity — a class effect of MMAF-based ADCs.

Metric	Value
Total FAERS Reports	290
Serious Outcomes	243 (83.8%)
Non-Serious	43 (14.8%)
Unknown/Missing	4 (1.4%)

Top 10 Reported Adverse Reactions

Reaction	Count	% of Total
DEATH	44	15.2%
OCULAR TOXICITY	24	8.3%
PLASMA CELL MYELOMA	23	7.9%
KERATOPATHY	22	7.6%
VISUAL ACUITY REDUCED	19	6.6%
VISUAL IMPAIRMENT	18	6.2%
VISION BLURRED	16	5.5%
CATARACT	14	4.8%
CORNEAL EPITHELIAL MICROCYSTS	13	4.5%
DRY EYE	13	4.5%

Source: openFDA FAERS (search: patient.drug.medicinalproduct:Blenrep), accessed 2026-05-26. NOTE: FAERS data consists of unvalidated spontaneous reports; counts reflect reporting volume, not incidence rates.

> **Ocular Toxicity — The Achilles' Heel of MMAF ADCs:** Blenrep's dose-limiting keratopathy (corneal microcystic epithelial changes) affected ~70% of patients in clinical trials, with ~40% experiencing grade ≥2 visual acuity reduction. This required frequent ophthalmological monitoring and dose holds/delays, severely limiting real-world utility. HDP-101's amanitin payload (RNA Pol II inhibitor) operates via an entirely different mechanism — it is not a tubulin inhibitor and may not cause the same corneal toxicity. However, amanitin's own toxicity profile (primarily hepatic) must be carefully managed.

5.2 HDP-101 vs. Blenrep — Head-to-Head Comparison

Parameter	Blenrep (Belantamab)	HDP-101
Target	BCMA (CD269)	BCMA (CD269)
Antibody Type	Afucosylated humanized IgG1	Humanized monoclonal
Payload	MMAF (monomethyl auristatin F) — microtubule inhibitor	α-Amanitin — RNA polymerase II inhibitor
Mechanism	Disrupts microtubule dynamics → mitotic arrest	Inhibits transcription → cell death in dividing AND resting cells
Drug-to-Antibody Ratio	~4	Not publicly disclosed

Primary Toxicity	Ocular (keratopathy, visual acuity loss)	Expected: hepatic (amanitin class effect)
Clinical Phase	Approved (combo); mono withdrawn	Phase 1/2 (NCT04879043)
Key Advantage	Validated BCMA ADC concept	Novel MOA; may address resting/resistant cells

Source: ClinicalTrials.gov (NCT04879043), EMA register (Blenrep), PubMed (Strassz et al. Blood 2020), accessed 2026-05-26

5.3 ADC Pipeline in Multiple Myeloma

Beyond Blenrep and HDP-101, several other BCMA-directed ADCs have been explored in clinical development. Notable programs include:

- **AMG 224** (Amgen) — Anti-BCMA ADC with maytansinoid payload (DM4). Phase 1 completed; development discontinued due to lack of differentiation.
- **MEDI2228** (AstraZeneca/MedImmune) — Anti-BCMA ADC with PBD dimer payload. Phase 1 completed; development discontinued.
- **HDP-101** (Heidelberg Pharma) — The only BCMA ADC in active clinical development with a non-tubulin payload.

The failure of early BCMA ADCs (AMG 224, MEDI2228) and Blenrep's monotherapy withdrawal highlights the challenges in BCMA ADC development. However, HDP-101's novel payload mechanism may provide a therapeutic window not achievable with tubulin- or PBD-based ADCs.

Source: ClinicalTrials.gov; published literature. Accessed 2026-05-26

6. Cross-Modality BCMA Competition

6.1 The BCMA-Targeted Therapy Ecosystem

BCMA (B-cell maturation antigen) is the dominant target for novel multiple myeloma therapies. Seven BCMA-directed therapies are now approved in the EU across three modalities: ADCs, bispecific antibodies, and CAR-T cell therapies.

6.2 Head-to-Head Modality Comparison

Modality	Drug	Sponsor	Mechanism	EU Approval (Line)	Key Limitation
ADC	Blenrep	GSK	BCMA-directed, MMAF payload	Authorised, 1L+ (combo only)	Ocular toxicity; mono withdrawn
ADC (investigational)	HDP-101	Heidelberg Pharma	BCMA-directed, amanitin payload	Phase 1/2 (not yet approved)	Clinical data pending; hepatic toxicity risk
Bispecific	Tecvayli	JNJ/ Janssen	BCMAxCD3 T-cell engager	Conditional, 3L+	CRS/ICANS; requires step-up dosing; IV/SC
Bispecific	Elrexfio	Pfizer	BCMAxCD3 T-cell engager	Conditional, 3L+	CRS/ICANS; infection risk
Bispecific	Lynozyfic	Regeneron	BCMAxCD3 T-cell engager	Conditional, 3L+	CRS/ICANS; infection risk
Bispecific (GPRC5D)	Talvey	JNJ/ Janssen	GPRC5DxCD3 T-cell engager	Conditional (orphan), 3L+	Different target; dysgeusia, nail toxicity
CAR-T	Abecma	BMS	BCMA CAR-T (ide-cel)	Authorised (orphan), **2L+**	Manufacturing complexity; CRS/ ICANS; lymphodepletion
CAR-T	Carvykti	JNJ/ Janssen	BCMA CAR-T (cilta-cel)	Authorised (orphan), **1L+**	Manufacturing complexity; CRS; neurotoxicity; T-cell exhaustion risk

Source: EMA Human Medicines Register, accessed 2026-05-26

6.3 Strategic Positioning by Line of Therapy

Key Insight — Treatment Sequence Matters: The approved BCMA therapies occupy different lines of therapy:

- **1L+:** Carvykti (lenalidomide-refractory), Blenrep (combo)
- **2L+:** Abecma
- **3L+:** Tecvayli, Talvey, Elrexfio, Lynozyfic
- **HDP-101 opportunity:** Phase 2a excludes prior BCMA therapy — positions for BCMA-naïve patients. This could be in 2L+ or 3L+ BCMA-naïve space

6.4 Differentiating Factors for HDP-101

Advantages vs. Bispecifics

- No cytokine release syndrome (CRS) risk (no T-cell engagement)
- No step-up dosing required (standard IV infusion)
- No immune effector cell-associated neurotoxicity (ICANS)
- Potentially suitable for outpatient administration
- No T-cell exhaustion concerns
- Conventional manufacturing (no cell therapy complexity)

Advantages vs. CAR-T

- Off-the-shelf product — no manufacturing delay (vs. 3-6 weeks for CAR-T)
- No lymphodepleting chemotherapy required
- No CRS/ICANS risk
- Lower cost of goods (estimated)
- No apheresis requirement
- Suitable for patients with inadequate T-cell function

⚠️ **HDP-101's Key Risks:**

- **No clinical efficacy data yet published.** The Phase 1/2 trial (NCT04879043) has not reported results. All competitive claims rest on preclinical data.
- **Amanitin payload carries hepatic toxicity risk.** α-Amanitin is a potent hepatotoxin in its native form. The ADC format is designed to mitigate this, but liver monitoring will be essential.
- **Blenrep's failure demonstrates the difficulty of BCMA ADC monotherapy.** Even with a different payload, single-agent activity may be limited.
- **Competing against well-established modalities.** Bispecifics and CAR-Ts have generated robust efficacy data with ORR of 60-70%+ in heavily pretreated populations.

7. Strategic Positioning Analysis

7.1 HDP-101's Potential Positioning in the MM Landscape

Based on the verified competitive landscape data, HDP-101 has several potential positioning strategies for Dr. Liu's consideration:

Option A: BCMA-Naïve, Early-Line (2L+) — Preferred Strategy

- **Rationale:** Phase 2a's BCMA-exclusion criterion uniquely positions HDP-101 for patients who have not received prior BCMA therapy. This is the only BCMA-directed agent with this specific enrollment criterion.
- **Market Need:** As BCMA bispecifics and CAR-Ts move into earlier lines (Carvykti already 1L+), an entire population of patients will need BCMA-naïve options.
- **Competitive Window:** Patients typically receive 2-3 prior lines before BCMA-targeted therapy. HDP-101 could position as a BCMA-directed option *before* bispecific or CAR-T.
- **Risk:** Requires demonstration of compelling efficacy versus IMiD/PI/mAb combinations.

Option B: Post-IMiD/PI/mAb, Pre-Bispecific/CAR-T (3L+)

- **Rationale:** Patients who have exhausted IMiDs, PIs, and anti-CD38 mAbs but have not yet received BCMA-directed therapy represent a large addressable population.
- **Market Size:** Estimated ~15,000-25,000 patients/year in EU5+US who are triple-class exposed and BCMA-naïve.
- **Risk:** This space is increasingly contested as bispecifics move earlier.

Option C: Combination Strategy

- **Rationale:** Following Blenrep's precedent (combination with bortezomib/dexamethasone and pomalidomide/dexamethasone), HDP-101 could be developed in combination with standard MM backbones.
- **Opportunity:** The amanitin payload's different mechanism (RNA Pol II inhibition) may be synergistically combinable with IMiDs, PIs, and mAbs without overlapping toxicity.
- **Precedent:** No combination trials for HDP-101 are currently registered on ClinicalTrials.gov, representing a gap to explore.

7.2 Recommended Near-Term Actions

Priority	Action	Rationale	Timeline
🔴 **1**	Prepare for China Phase 1 primary completion readout (30 June 2026)	DLT data from the China dose-escalation cohort will inform global RP2D and potentially accelerate enrollment	Immediate (next 5 weeks)

🔴 **2**	Advocate for Phase 2a data disclosure (NCT04879043 primary completion Aug 2025 — data expected imminently)	Primary completion was August 2025; study completion May 2026. Initial efficacy data (ORR) should be available	Q2-Q3 2026
🟡 **3**	Evaluate combination therapy development path	Blenrep precedent shows combinations unlock broader labels. Pomalidomide or daratumumab combinations could differentiate	Q3-Q4 2026
🟡 **4**	Develop regulatory strategy for EU/US BCMA-naïve indication	Phase 2a's BCMA-exclusion design supports a differentiated label. Early EMA/FDA engagement recommended	Q3 2026
🟢 **5**	Strengthen IP position around amanitin-ADC platform	If HDP-101 demonstrates clinical proof-of-concept, the platform value increases significantly for other indications	Ongoing

7.3 Market Context

The global multiple myeloma therapeutics market is estimated to be worth approximately ~$20-25B annually (2025 estimate) and growing at an estimated CAGR of 8-10%. Key market segment estimates:

- **CAR-T cell therapy market:** Estimated ~$5B by 2026 (Abecma + Carvykti + emerging products)
- **Bispecific antibody market:** Estimated ~$3-4B by 2026-2027 (Tecvayli, Talvey, Elrexfio, Lynozyfic)
- **ADC market in MM:** Currently limited to Blenrep (combo only) — estimated ~$200-400M globally. Significant upside potential if a well-tolerated BCMA ADC enters the market
- **Total addressable MM ADC market:** Estimated ~$2-4B if a first-line-compatible BCMA ADC with manageable safety profile is developed

Source: All figures in this section are author's estimates based on publicly available market analyses and should be treated as approximations.

8. Regulatory Outlook

8.1 EU Regulatory Pathway for HDP-101

Based on the current EMA landscape for multiple myeloma drugs, HDP-101 could pursue several regulatory pathways:

Orphan Designation

Multiple myeloma qualifies as a condition for which EU orphan designation may be granted (prevalence <5 in 10,000). Several MM drugs have received orphan designation in the EU (Darzalex, Abecma, Carvykti, Talvey). HDP-101 may be eligible for orphan designation, which would provide 10 years of market exclusivity upon approval, fee reductions, and protocol assistance.

Conditional Approval Pathway

Four of the four approved bispecific antibodies for MM (Tecvayli, Talvey, Elrexfio, Lynozyfic) received conditional approval in the EU, reflecting EMA's willingness to use this pathway for MM drugs that address unmet medical need. HDP-101, if it demonstrates promising Phase 2 data, could qualify for conditional approval — though the bar for a monotherapy BCMA ADC after Blenrep's withdrawal may be higher.

PRIME Designation Potential

EMA's PRIME (Priority Medicines) scheme provides enhanced support for medicines targeting an unmet medical need. HDP-101's novel mechanism (first RNA Pol II inhibitor ADC) could qualify, providing early dialogue and accelerated assessment.

8.2 US FDA Pathway

While not the primary focus of this EU-centric report, the US market would be critical for HDP-101's commercial success. Potential pathways include:

- **Fast Track Designation:** Available for drugs addressing unmet need in serious conditions
- **Breakthrough Therapy Designation:** Requires preliminary clinical evidence of substantial improvement over existing therapies
- **Accelerated Approval (Surrogate Endpoint):** Based on ORR — the primary endpoint of Phase 2a — which is an established surrogate endpoint in MM

8.3 China Regulatory Pathway (NMPA)

The China Phase 1 trial (NCT07529782), sponsored by Hangzhou Zhongmei Huadong Pharmaceutical, is the first step toward NMPA registration. The Chinese regulatory environment for innovative oncology drugs has become more favorable, with:

- Priority review for innovative drugs addressing unmet medical needs
- Acceptance of foreign clinical data under certain conditions (especially for orphan diseases)
- Potential for conditional approval based on Phase 2 data with surrogate endpoints

Source: EMA regulatory framework; ClinicalTrials.gov; NMPA regulations. Accessed 2026-05-26.

9. Key Takeaways for Dr. Dongzhou Liu

1. HDP-101's Novel Mechanism is Its Greatest Asset and Greatest Unknown.
The α-amanitin payload (RNA Pol II inhibition) is a genuinely first-in-class ADC mechanism. It could avoid the ocular toxicity that doomed Blenrep and potentially address resting/resistant myeloma cells. However, no clinical efficacy data has been published. The next 6 months — with data from both the global (NCT04879043) and China (NCT07529782) trials — will be defining.

2. The BCMA-Naïve Positioning is a Potent Strategic Advantage.
HDP-101's Phase 2a BCMA-exclusion criterion is unique among all BCMA-directed agents in development. This positions HDP-101 for patients who have not yet received BCMA-targeted therapy — potentially in earlier lines (2L-3L) before bispecifics or CAR-Ts. This could be a significant first-mover advantage if efficacy is demonstrated.

3. The China Partnership is a Significant Asset.
The collaboration with Hangzhou Zhongmei Huadong Pharmaceutical provides non-dilutive funding, Chinese market access, and regulatory expertise. The imminent primary completion of the China Phase 1 (30 June 2026) is the most important near-term catalyst for the company.

4. Combination Strategy Should Be Explored Early.
Blenrep's trajectory — from failed monotherapy to viable combination regimens — demonstrates that BCMA ADCs may require combination partners. Preclinical and early clinical exploration of HDP-101 combinations with pomalidomide, daratumumab, or carfilzomib could open a broader label and differentiate from bispecifics.

5. Financial Position Requires Strategic Capital Allocation.
With a market capitalization of ~€128M and enterprise value of ~€120.5M, Heidelberg Pharma is a small-cap biotech with a single lead program. The float of ~10.1M shares (21.6% of outstanding) suggests limited liquidity. Near-term value inflection points (China data, Phase 2a ORR data, potential partnership expansion) will be critical for shareholder value.

6. The MM Competitive Landscape is Crowded but Differentiable.
Fifteen-plus drugs are approved for MM in the EU across 7+ mechanisms of action. Four BCMA bispecifics and two BCMA CAR-Ts compete in the late-line space. However, no approved therapy combines BCMA targeting with an RNA Pol II inhibitor mechanism. If HDP-101 can demonstrate a differentiated safety profile (no ocular toxicity, no CRS/ICANS, manageable hepatic safety) with meaningful efficacy, it occupies a unique niche.

7. Regulatory Pathways Exist for Accelerated Development.
The precedent of conditional approvals for all four MM bispecifics demonstrates EMA's willingness to use accelerated pathways. Orphan designation, PRIME eligibility, and potential FDA Breakthrough designation could all be pursued to accelerate HDP-101's development.